BEFORE YOU READ

Six functions in three pairs

The six trigonometric functions pair up by behaviour: one pair is bounded, two pairs are unbounded.

- L0 **The six trigonometric functions**
  - L1 **Sine and cosine** bounded and continuous
    - L2 **sin** y-coordinate; odd — period 2π
    - L2 **cos** x-coordinate; even — period 2π
  - L1 **Tangent and cotangent** unbounded, half-period π
    - L2 **tan** odd — $\sin/\cos$
    - L2 **cot** odd — $\cos/\sin$
  - L1 **Secant and cosecant** unbounded outside $[-1, 1]$
    - L2 **sec** even — $1/\cos$
    - L2 **csc** odd — $1/\sin$

TAKEAWAY

Learn the pairs

Sine and cosine are bounded and continuous; tangent and cotangent are unbounded with half-period π; secant and cosecant are unbounded outside $[-1, 1]$.